

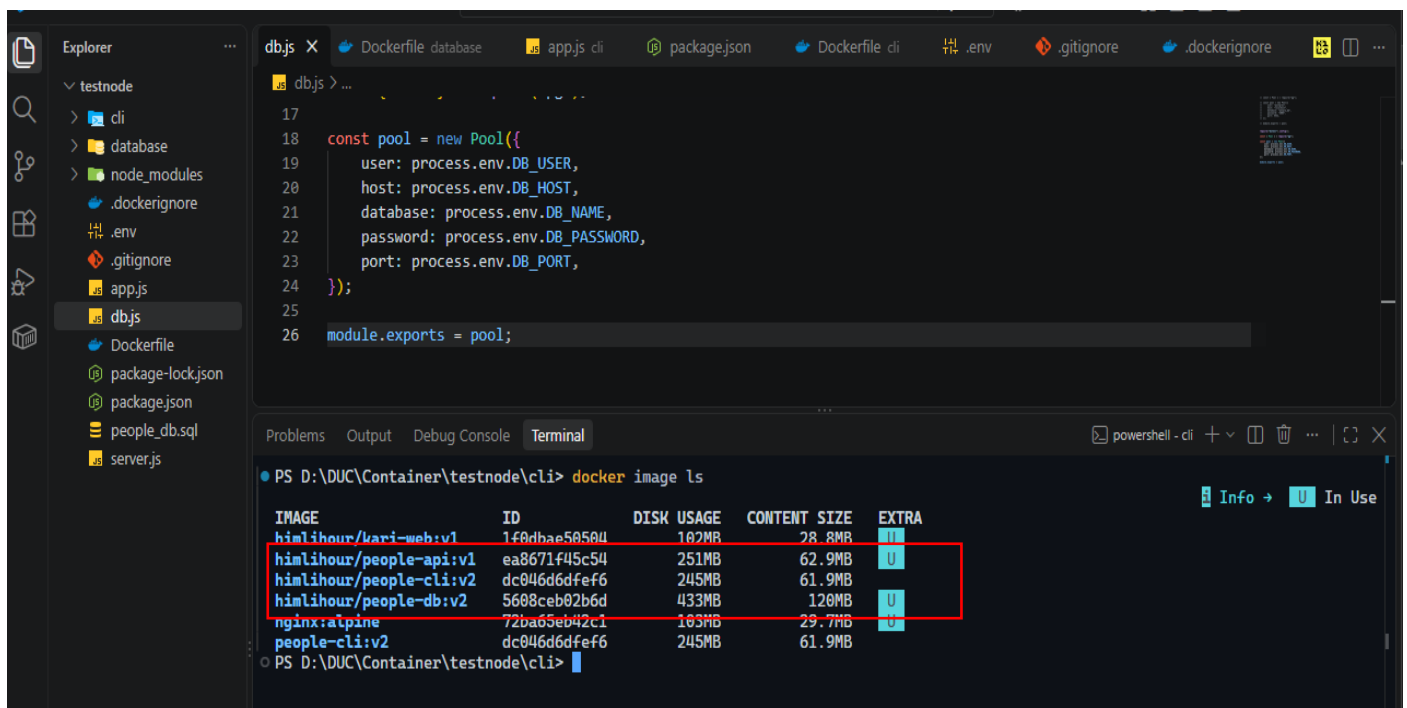
ហ៊ីម លីហ្គឺរ

ថ្នាក់ PGA

DUC20240776

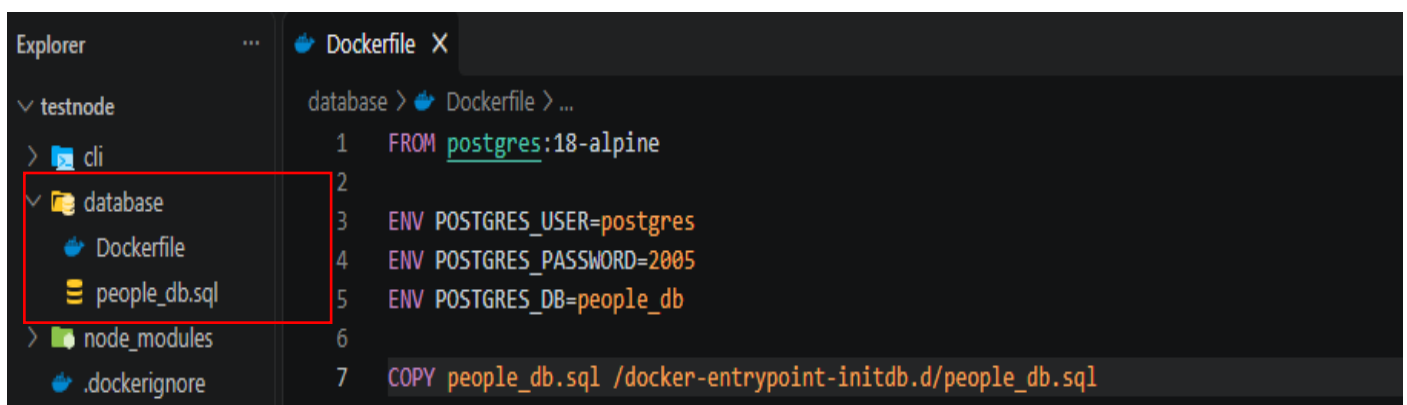
Homework :

បង្កើត Docker Images សម្រាប់ប្រព័ន្ធ Backend ហើយអាចភ្ជាប់ទៅ Database បាន។ Project នេះ មាន Docker Images ចំនួន 3 គឺ CLI, API និង Database។ Images ទាំងនេះត្រូវបាន Push ទៅ Docker Hub ហើយអាច Pull មកប្រើនៅលើ Computer ផ្សេងទៀតបាន



មាន 3 Image

+Image : himlihour/people-db:v2 (database Postgres)



ដើម្បីបង្កើត Image : `cd database`

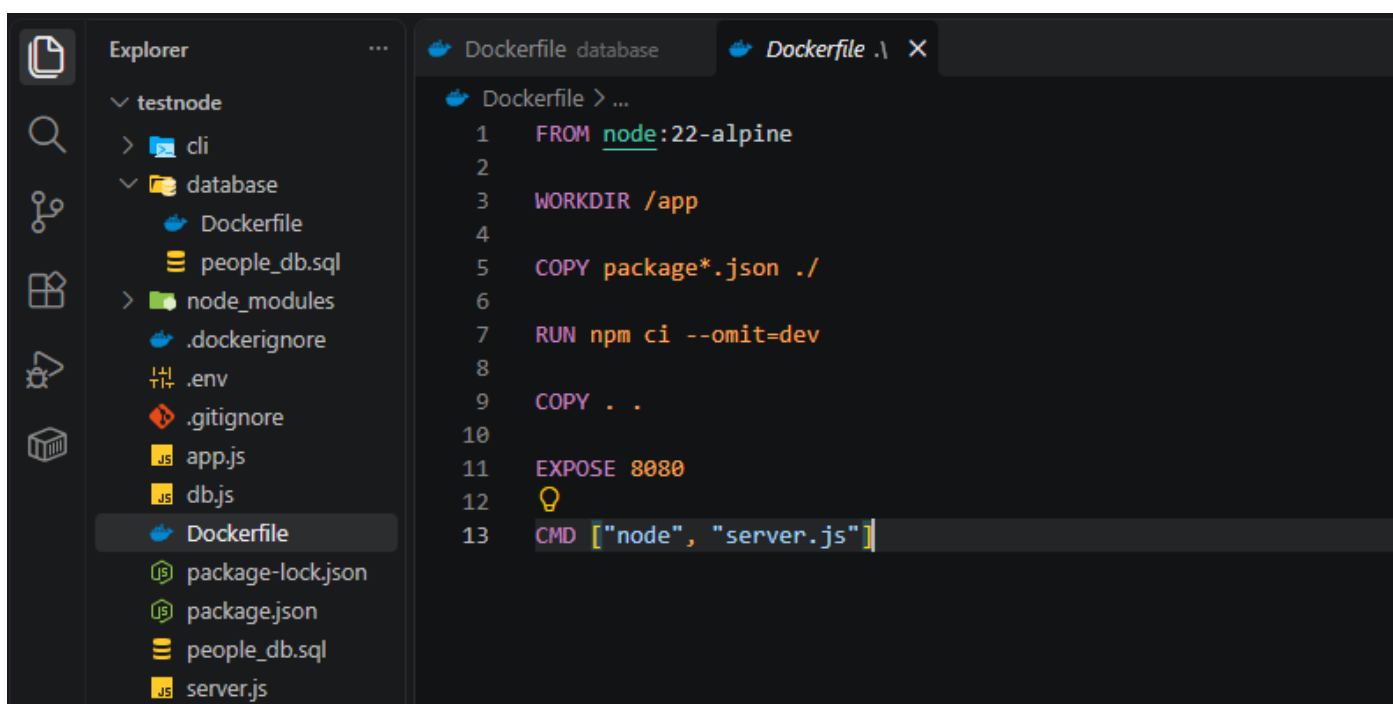
Run : `docker build -t people-db:v2 .` (ប្រើដើម្បីបង្កើត Docker Image ពី Dockerfile និងឯកសារនៅក្នុង Folder បច្ចុប្បន្ន។ Image នេះមានឈ្មោះ people-db និង Tag v2

Then : `docker tag people-db:v2 himlihour/people-db:v2` (ប្រើដើម្បីដាក់ឈ្មោះ Docker Hub ទៅឱ្យ Image ដែលបានបង្កើត ដើម្បីអាច Push ទៅ Docker Hub បាន។

`docker push himlihour/people-db:v2` (ប្រើដើម្បីបញ្ជូន people-db:v2 ពី Computer របស់យើងទៅ Docker Hub ដើម្បីឱ្យ Computer ផ្សេងទៀតអាច Download បាន

ទាញយក : `docker pull himlihour/people-db:v2`

+Image : himlihour/people-api:v1 (backend api)



ដើម្បីបង្កើត Image :

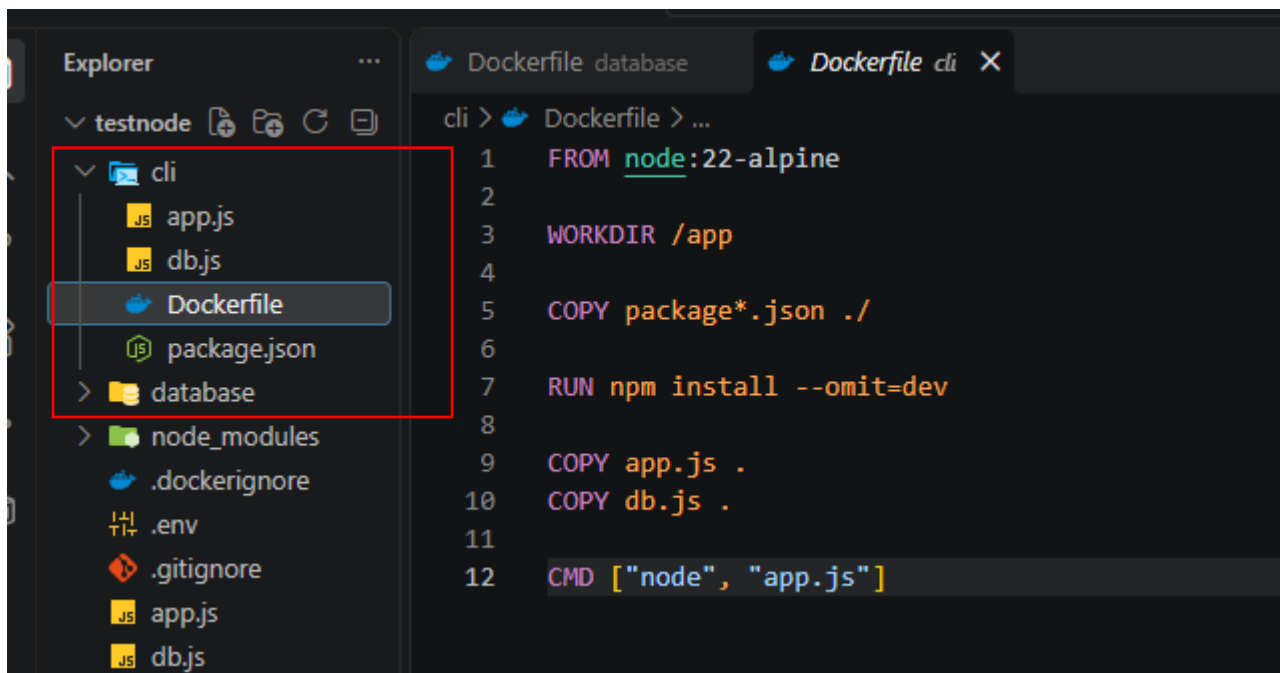
```
docker build -t people-api:v1 .
```

```
docker tag people-api:v1 himlihour/people-api:v1 docker
```

```
push himlihour/people-api:v1
```

ទាញយក : `docker pull himlihour/people-api:v1`

+Image : himlihour/people-cli:v2 (user interface)



ដើម្បីបង្កើត Image : `cd cli`

```
docker build -t people-cli:v2 .
```

```
docker tag people-cli:v2 himlihour/people-cli:v2
```

```
docker push himlihour/people-cli:v2
```

ទាញយក docker pull himlihour/people-cli:v2

+ របៀបតេស្តលើ OS ផ្សេង

docker pull himlihour/people-cli:v2

docker pull himlihour/people-api:v1

docker pull himlihour/people-db:v2

```
Status: Downloaded newer image for himlihour/people-cli:v2
docker.io/himlihour/people-cli:v2
lihour@debian:~$ docker image ls
REPOSITORY          TAG          IMAGE ID      CREATED      SIZE
himlihour/people-cli v2           02e8889cd122 3 hours ago  171MB
himlihour/people-db  v2           aefa8d1a5407 6 hours ago  304MB
himlihour/people-api v1           c3e90757e9fb 7 hours ago  172MB
alpine               latest       320994c3b997 7 days ago   8.42MB
lihour@debian:~$ |
```

```
lihour@debian:~$ docker network create people-network
5bf3cf311b4cf38c4f3a92114e8f7c879ca05a307f9a4f3099f884fce367d691
lihour@debian:~$ docker network ls
NETWORK ID          NAME                DRIVER          SCOPE
635c3d7ef663       bridge              bridge          local
296fc9062051       host                host            local
bd3367cc97cd       none                null            local
5bf3cf311b4c       people-network      bridge          local
lihour@debian:~$ |
```

docker network create people-network : Creates a private Docker network so the containers can communicate with each other using container names.

docker network ls : check llist

Start the database

```
docker run -d \

--name people-db \

--network people-network \
```

```
lihour@debian:~$ docker run -d \
  --name people-db \
  --network people-network \
  himlihour/people-db:v2
783393fe811b7c70c41b95d0fdab307301001cc74ebbc613d701e062163be813
lihour@debian:~$ |
```

[docker network create people-network](#) : Creates a private Docker network so the containers

Start the API

```
docker run -d \

  --name people-api \

  --network people-network \

  -p 8080:8080 \

  -e DB_USER=postgres \

  -e DB_HOST=people-db \

  -e DB_NAME=people_db \

  -e DB_PASSWORD=2005 \

  -e DB_PORT=5432 \

  himlihour/people-api:v1
```

```
lihour@debian:~$ docker run -d \
  --name people-api \
  --network people-network \
  -p 8080:8080 \
  -e DB_USER=postgres \
  -e DB_HOST=people-db \
  -e DB_NAME=people_db \
  -e DB_PASSWORD=2005 \
  -e DB_PORT=5432 \
  himlihour/people-api:v1
4bb50385898c3da734ade38814264090df1c6fd387eaf82ab7e98f7b9b9ad13b
lihour@debian:~$ |
```

```

4bb50385898c3da734ade38814264090df1c6fd387eaf82ab7e98f7b9b9ad13b
lihour@debian:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PO
RTS
4bb50385898c   himlihour/people-api:v1            "docker-entrypoint.s..." 59 seconds ago Up 58 seconds 0.
0.0.0:8080->8080/tcp, :::8080->8080/tcp   people-api
783393fe811b   himlihour/people-db:v2            "docker-entrypoint.s..." About a minute ago Up About a minute 54
32/tcp                                     people-db
lihour@debian:~$ curl http://localhost:8080/people
[{"id":1,"name":"kari","age":21},{ "id":3,"name":"leang chhay","age":22},{ "id":4,"name":"karin","age":23},{ "id
":5,"name":"lina","age":22},{ "id":6,"name":"mara","age":12},{ "id":7,"name":"july","age":12},{ "id":9,"name":"H
lihour@debian:~$ |

```

Start the CLI

```

docker run -it --rm \

--name people-cli \

--network people-network \

-e DB_USER=postgres \

-e DB_HOST=people-db \

-e DB_NAME=people_db \

-e DB_PASSWORD=2005 \

-e DB_PORT=5432 \

himlihour/people-cli:v2

```

PEOPLE MANAGEMENT SYSTEM	
1. Add person	4. Update person
2. View people	5. Delete person
3. Search person	6. Statistics
7. Exit	
Enter your choice:	

PEOPLE LIST		
ID	NAME	AGE
1	kari	21
3	leang chhay	22
4	karin	23
5	lina	22
6	mara	12
7	july	12
9	Hira	12
10	Kong	12
11	hongdary	34
12	dara	23

Press Enter to continue...|